COMPUTER NETWORKS

Week-16

Determining What to Measure

- Before any measurements can take place one must determine *what* to measure

Points to note..

- Network Provider
  - Is the design meeting requirements for various traffic classes or applications?
  - How can I demonstrate the superior performance of my service offering?
- Customer
  - Is Network Performance $\geq$ Agreement?
- 3rd Parties
  - What does "the net" look like? Hot spots?
  - What Network Provider is "best"?

Key metrics for network performance evaluation

- Latency
- Throughput
- Response time
- Arrival rate
- Utilization
- Bandwidth
- Loss
- Routing
- Reliability

Four major concepts

- Latency (travel time)
- Throughput (actual transfer rate)
- Packet loss
- Jitter (delay variation)

Bandwidth

- The first, bandwidth in hertz, refers to the range of frequencies in a composite signal or the range of frequencies that a channel can pass.
- The second, bandwidth in bits per second, refers to the speed of bit transmission in a channel or link. Often referred to as Capacity.

Bandwidth

- Hertz measures the frequency of a signal or the width of a communication channel
- How much of frequency spectrum is available

- Used for analog communication channels (Traditional landline telephone, FM Radio broadcast,, older analog TVs) , frequency range, wireless spectrum (e.g. bandwidth of 20MHz), channel capacity in terms of frequencies

Bandwidth

- bps measures amount of digital data transmitted per second
- Internet speed, data communication rate, network throughput, download speed
- Upload speed etc.

- If a link is 100Mbps → it can transmit 100 million bits/second

Bandwidth

- A communication channel with more frequency bandwidth (Hz) can usually carry more data (bps)
- In computer networks, the term “bandwidth” is commonly used informally for transmission speed

Measuring bandwidth

Unit of Bandwidth	Abbreviation	Equivalence
Bits per second	bps	1 bps = fundamental unit of bandwidth
Kilobits per second	kbps	1 kbps = ~1,000 bps = 10^3 bps
Megabits per second	Mbps	1 Mbps = ~1,000,000 bps = 10^6 bps
Gigabits per second	Gbps	1 Gbps = ~1,000,000,000 bps = 10^9 bps
Terabits per second	Tbps	1 Tbps = ~1,000,000,000,000 bps = 10^{12} bps

Network Performance

- Two characteristics:
- Delay
- Throughput

Delay

- How long does it take for a bit of data to travel across the network from one compute to the other
- It is measured in seconds or fractions of seconds

Types of delay

- Propagation Delay→ time to travel across a medium (time for signal to travel)
- Switching Delay → It is the time required for network component (hub, bridge, packet switch) to forward data
- Access Delay → It is the time required to get control of medium
- It occurs because multiple devices may be trying to use the same medium at the same time
- → waiting time before transmission begins

Types of delay

- Queuing Delay → Time required in packet switches→ waiting in device buffer

- Processing Delay → Time to process packet

- Access delay is noticeable when crowded Wi-Fi networks often feel slow even with good bandwidth

Ex-1

- Assume we need to download text documents at the rate of 100 pages per sec. What is the required bit rate of the channel (page is an average of 24 lines with 80 characters in each line)?
- Solution
- If we assume that one character requires 8 bits (ascii), the bit rate is

$$\boxed{100 \times 24 \times 80 \times 8 = 1{,}636{,}000 \text{ bps} = 1.636 \text{ Mbps}}$$

$$\text{Mbps} = \frac{\text{bits per second (bps)}}{1,000,000}$$

1Mbps is the rate of transfer of one million bits per second

Ex-2

- A digitized voice channel is made by digitizing a 4-kHz (1 kilo hertz = 1000 Hz) bandwidth analog voice signal. We need to sample the signal at twice the highest frequency (two samples per hertz). We assume that each sample requires 8 bits (bit depth). What is the required bit rate?
  Solution
- The bit rate can be calculated as

$$\boxed{2 \times 4000 \times 8 = 64{,}000 \text{ bps} = 64 \text{ kbps}}$$

kHz measures frequency (how many times per second a sound wave oscillates)
bps → kbps (divide by 1000)

Throughput

- The rate at which data can be sent through the network
- The throughput capability of the underlying hardware is called bandwidth
- Speed can be a synonym for throughput

Throughput

- If a packet switch has a queue of packets waiting when a new packet arrives
- The new packet will be placed on the entire queue and will need to wait while the switch forwards the previous packets
- Throughput and delay are not completely independent
- As traffic in a computer network increase, delays increase a network that operates at close to 100% of its throughput capacity experiences severe delay

What is Packet Reordering?

- Packets arrive at Dst, but not in send order.
  - 1, 2, 3, 7, 8, 9, 10, 11,... Loss, no reordering
  - 1, 2, 3, 7, 8, 9, 4, 5, 6, 10, 11,...reordering
- In the "world of order" all these packets are of interest.
  - 1, 2, 3, 7, 8, 9, 4, 5, 6, 10, 11,...
  - | Early | Late |
  - No reordering until Late Packets Arrive
  - # of Early Packets => Reordering Extent

What is Packet Reordering?

- Receivers must perform work to restore order

1, 2, 3, 7, 8, 9,10, 4, 5, 6, 11, 12,...

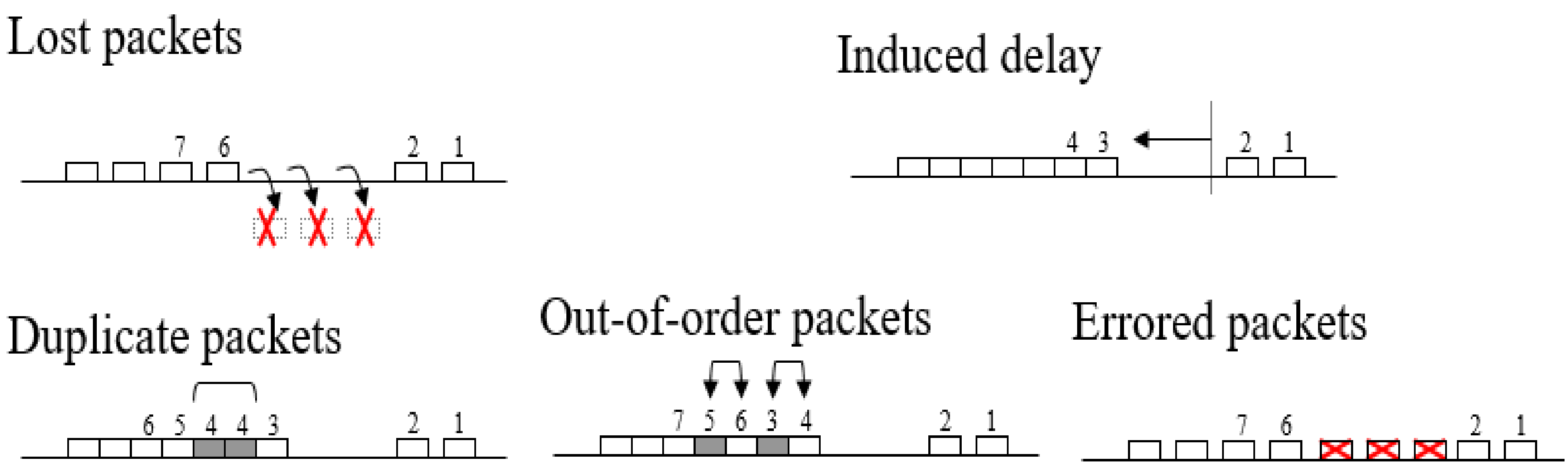
| Buffered | |Reordered|

Reordered packet

- Packet n is designated reordered when its sequence number is less than the Next Expected threshold (set by the arrival of a previous packet).

Failure recovery time

- Five possible recovery scenarios:



Network Performance

- Effective network performance → improved user satisfaction
- Network performance is important for:
- Employees in an organization
- E-commerce website customers
- Clients of a financial institution
- Users of educational content

Network Performance/efficiency

- Network efficiency refers to how effectively a computer network utilizes its available resources to transfer data from source to destination
- Delivers data quickly
- Uses bandwidth effectively
- Minimizes delays and errors
- Provides reliable communication

Network Performance

- Performance monitoring on an ongoing basis → ensures network quality
- Network performance measurement tools can be broadly categorized into two types:
- Passive
- Active

Network Performance

- Network Traffic Measurement
- Measuring the amount & type of traffic on a particular network
- → effective bandwidth management

Network Performance

- Network Monitoring
- A system that constantly monitors a network for slow or failing system
- Notifies the network administrator about the availability
- → percentage of a specified time interval during which the system was available for normal use
- Network administrator also keeps an eye on link utilization
- → throughput for the link expressed as a percentage of the access rate

Network Performance

- Passive network measurement tools monitor (or measure) existing applications on the network to gather data on performance metrics
- Ensures no network disruption
- No additional traffic is introduced for such monitoring
- A realistic assessment of the user experience may be obtained
- Used to perform link usage, throughput, intrusion detection
- Processing overhead (in terms of CPU utilization)

Network Performance

- Active networking performance measurement tools require an additive level of data traffic
- These must be scheduled appropriately to minimize impact on existing network traffic
- Performed by sending test traffic into the network
- Generate test packets periodically or on-demand
- Measure the performance of test packets → statistics
- Test packets can be blocked by firewall or processed at low priority by routers
- Checks delay, packet loss and network availability

Network Performance

- **Latency**
- The amount of time it takes for data to travel from one defined location to another (sender to receiver)
- Also referred to as delay (unit : milliseconds : ms)
- Ideally, the target latency of a network is as close to zero as possible
- Packet queuing in switched networks and the refractive index of fiber optic cabling are examples of variables that can increase latency

Latency

- Example-1
- If a packet takes 40 ms to reach the destination
- Latency = 40 ms
- Video conferencing requires low latency
- Online gaming requires very low latency

Network Performance

- **Packet Loss**
- Number of packets transmitted from one destination to another that fail to transmit
- This can be quantified by capturing traffic data on both ends, then identifying missing packets and/or retransmission of packets
- Can be caused by network congestion, router performance and software issues

Network Performance

- Can be detected by users in the form of voice and streaming interruptions or incomplete transmission of files
- Retransmission is a method utilized by network protocols to compensate for packet loss
- → can aggravate network congestion by the increased traffic volume caused by retransmission

Network Performance

- Response time analysis can be done to identify the root cause of the issue
- A review of response time for TCP connections can pinpoint which applications are contributing to the bottleneck

Packet Loss

- Packet loss occurs when packets fail to reach their destination
- Example-2
- Packets Sent = 1000
- Packets Received = 980
- Packet Loss Rate= $[(1000-980)\ /\ 1000\ ]\times 100$
  $=2\%$
- Packet Loss = 2%

Network Performance

- **Throughput**
- Commonly defined as the amount of material or items passing through a particular system or process
- For network performance measurement, throughput is defined in terms of the amount of data or number of data packets that can be delivered in a pre-defined time frame

Throughput

- Actual amount of useful data successfully delivered
- Example-3
- Bandwidth = 100 Mbps
- Actual data transferred = 75 Mbps
- Throughput = 75 Mbps
- Since throughput is lower than bandwidth, some capacity is being lost due to overhead or congestion

Network Performance

- **Bandwidth**
- Usually measured in bits per second
- Amount of data that can be transferred over a given time period
- Bandwidth → a measure of capacity rather than speed
- For example, a bus may be capable of carrying 100 passengers (bandwidth), but the bus may actually only transport 85 passengers (throughput)
- A fiber connection with 1 Gbps bandwidth can theoretically transmit:
- 1,000,000,000 bits/sec

Network Performance

- **Jitter**
- Variation in time delay for the data packets sent over a network
- Jitter is the variation in packet arrival times
- Identified disruption in the normal sequencing of data packets
- Jitter is related to latency
- Jitter expresses itself in increased or uneven latency between data packets
- →can disrupt network performance → packet loss & network congestion
- Some level of jitter is to be expected and can usually be tolerated

Jitter

- Example-4
- Given Packet delays:
- Packet 1 = 20 ms
- Packet 2 = 25 ms
- Packet 3 = 18 ms
- Packet 4 = 30 ms
- Significant variations
- High jitter affects:
- Voice calls
- Video conferencing
- Live streaming

Network Performance

- **Latency vs Throughput**
- Throughput is a measurement of actual system performance → data transfer over a given time
- Latency is a measurement of the delay in transfer time
- Has direct impact on throughput
- Throughput is measured in units completed which is inherently influenced by latency in the system

Network Performance

- **Network Efficiency**
- Example-5
- Suppose a network link can transmit 100 Mbps, but only 80 Mbps of useful data reaches users because of protocol overhead and retransmissions
- Network Efficiency:
- (Useful Throughput/Available Bandwidth) x 100

- Network Efficiency = 80/100 x 100

  = 80%

Network Efficiency

- Example-6
- A network has:
- Bandwidth = 200 Mbps
- Throughput = 160 Mbps
- Calculate network efficiency

- Efficiency=[(160 / 200)]×100
  =80%
- Network Efficiency = 80%

Network Performance

Throughput vs useful throughput

Throughput : Total rate at which data is successfully transferred over the network

Useful Throughput : Rate at which actual user/application data is delivered

Useful Throughput (Goodput) = Throughput – Overhead – Retransmissions

Network Performance

- Throughput measures the total amount of data delivered across the network per second, including:
- Application data
- Protocol headers (TCP, IP, Ethernet)
- Acknowledgments
- Retransmissions

Network Performance

- Example-7
- Suppose a network transfers 100 Mbps of data
- This 100 Mbps includes:
- 80 Mbps user data
- 15 Mbps headers and control information
- 5 Mbps retransmitted packets
- Throughput = 100 Mbps

Network Performance

- Useful throughput measures only the data that is actually useful to the application or user
- It excludes:
- Protocol overhead
- Retransmissions
- Control packets

Network Performance

- Example-8
- Total throughput = 100 Mbps
- Headers = 15 Mbps
- Retransmissions = 5 Mbps
- Useful Throughput:
- 100 – 15 – 5 = 80 Mbps
- So..
- Throughput = 100 Mbps
- Useful Throughput = 80 Mbps

Ex-2

- **Downloading a File**
- Example-9
- Network bandwidth = 1 Gbps
- Measured throughput = 900 Mbps
- TCP/IP overhead = 50 Mbps
- Retransmissions = 30 Mbps
- Useful Throughput:
- 900−50−30=820 Mbp
- The user effectively receives data at 820 Mbps, even though the network is carrying 900 Mbps

Useful Throughput

- Streaming video quality depends on useful throughput
- Cloud file uploads depend on useful throughput
- Web page loading speed depends on useful throughput

Challenges

- Network performance measurement solutions should be designed with the user in mind
- Slight degradation/latency may be fine
- Finding these acceptable limits is the key to establishing relevant testing and monitoring
- Packet shaping is a method used to prioritize package delivery for different applications
- →adequate bandwidth is consistently allocated to the most important categories
- File compression is another innovation that decreases the bandwidth and memory consumed

Challenges

- Implementation of effective network performance measurement practices
- If problems with servers, routing, delivery or bandwidth can be detected in real time, practical solutions and preventative strategies are possible

Factors Affecting Network Efficiency

Congestion

- Too many devices compete for network resources

Protocol Overhead

- Headers, acknowledgments, and control messages consume bandwidth

Transmission Errors

- Corrupted packets require retransmission

Factors Affecting Network Efficiency

Distance

- Longer distances increase propagation delay

Network Hardware

- Old switches, routers, and cables can reduce performance

Security Mechanisms

- Encryption and inspection may add processing delay

What do we require?

Online Banking

- Low latency
- High reliability
- Minimal packet loss
- Even a small delay can affect transaction processing

What do we require?

Video Streaming

- High throughput
- Low jitter
- If throughput drops below the video bit rate, buffering occurs

What do we require?

Cloud Storage in Cloud Computing

- Sufficient bandwidth
- Low latency
- Reliable connectivity
- Poor network efficiency slows file synchronization and application performance

Emerging Trends

Emerging Trends

- Mobile adoption
- Experiment-driven decision making
- Business Process Automation

Mobile connectivity

- IT network allows employees to work from anywhere
- On average we are now working for 47 hrs per week
- No compromise on security

Why extended working hrs / week?

- Digital connectivity (smart phones, emails, messaging apps, remote work tools etc.
- Increased competition in global markets → hi productivity, faster response times expected etc.
- Knowledge-based jobs: work is measured by outcomes and not hours
- Hybrid & Remote work: extended working hours unintentionally
- Economic Pressures : hi living costs → overtime, freelance work, side business online

Experiment driven decisions

- Based on data collected by testing and retesting different variables
- → optimal results to be achieved
- Software today enable businesses today to base their decisions on real time changes
- → better innovation expected as businesses take risks with experiments (trying new things)

Business Process Automation (BPA)

- Networks are the backbone of automated business activities
- Automated systems continuously exchange data between applications, servers, databases, cloud platforms, and IoT devices → lots of data on networks
- Customer Relationship Management (CRMs)
- AI Chatbots
- Online transaction Systems
- → greater dependence on Cloud networks

Need of the day..

- Automated banking
- Online shopping
- Industrial robotics
- Logistic tracking
- → real-time decision making

Need of the day..

- Fiber optics networks
- 5G (higher speed, lower delay, better support for modern smart devices)
- Software Defined Networking (SDN)→ RIY about how control of network is separate from the physical network devices→ programmable, flexible network where network traffic is managed dynamically thru a central controller
- Edge computing

Edge Computing

- Data processing happens closer to where the data is generated, instead of sending everything to a central cloud or remote data center

- Data generated from a device (sensor, phone, camera, any machine) → processed by a nearby edge device (router, local server) → only important data is sent to the cloud

- Low latency
- Reduced bandwidth usage
- Real-time processing --. Real-time decisions

Edge Computing

- Used in:
- Smart cities → traffic light adjust based on real-time infor. of traffic flow
- Industrial automation→ quick fault detection in factories
- Autonomous vehicles → minimum human assistance → instant decisions on brakes, steer without waiting for cloud response
- Healthcare → wearable devices to monitor patients and generate alerts
- IoT Devices → sensors → process data locally → send summaries

What is required today?

- Measures against:
  - Cyberattacks
  - Ransomware
  - Unauthorized access

- Modern Networks rely on :
- Firewalls
- VPNs
- Encrypted communication
- Intrusion detection systems
- Zero trust security

Zero Trust Security

- No trust
- Continuous verification
- Least privilege access (minimum access granted to complete a job)
- Used in banking system, remote work, government systems, defense networks, cloud services